

Финансовая модель: 3 года

Версия 0.1 (2026-05-21). Рабочая финансовая модель для investor due diligence и board reporting. Все числа — на основе публичных данных и industry comparables.

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I. Обзор: три сценария

BEAR	BASE	BULL
Y1: \$100K ARR	Y1: \$250K ARR	Y1: \$500K ARR
Y2: \$600K ARR	Y2: \$2M ARR	Y2: \$5M ARR
Y3: \$2.5M ARR	Y3: \$12M ARR	Y3: \$30M ARR
Runway to Series A → Q3 2027 ← same across scenarios		
Pre-seed close: Q4 2026 (all scenarios)		

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II. Revenue projections

2.1. Customer count и ARPU

Три tier'a клиентов с различным ARPU и циклом продаж:

Customer Tiers:

Tier 1: Developer Teams (SMB)

Profile: engineering team 10-50 devs, SaaS / fintech / AI-native startup

ACV (Annual Contract Value): \$24K-\$60K (\$2K-\$5K/mec)

Sales cycle: 30-60 days

Churn rate: 15% annual (SMB is higher churn)

Gross margin: 75%

How won: self-serve + PLG motion (cpp-cli analyze → convert)

Tier 2: Enterprise

Profile: enterprise 100-500 devs, regulated industry / large tech

ACV: \$120K-\$360K (\$10K-\$30K/mec)

Sales cycle: 90-180 days

Churn rate: 7% annual

Gross margin: 70% (more customer success overhead)

How won: retroactive savings analysis → pilot → enterprise contract

Tier 3: Strategic / Sovereign AI Lab

Profile: G42, TII, Sber AI, Yandex — own LLM infrastructure

ACV: \$600K-\$2.4M (\$50K-\$200K/mec)

Sales cycle: 180-360 days (government procurement)

Churn rate: 5% annual

Gross margin: 65% (deep integration, dedicated support)

How won: government relationship + sovereign AI narrative

2.2. Base scenario: детальный год по году

Year 1 (2026): Pilot phase

YEAR 1 (2026) — BASE SCENARIO

Q3 2026 (Jul-Sep):

New Tier 1 customers: 2 (pilot conversions)
 New Tier 2 customers: 1 (first enterprise)
 New Tier 3 customers: 0
 Avg ACV: T1=\$3K/mec, T2=\$12K/mec
 ARR end of Q3: $2 \times \$36K + 1 \times \$144K = \$216K$

Q4 2026 (Oct-Dec):

New Tier 1: +1 (converted from retro analysis)
 New Tier 2: +1
 ARR end of Q4: $3 \times \$36K + 2 \times \$144K = \$396K \approx \$400K$

Year 1 Metrics:

Total customers EOY: 5
 ARR EOY: \$400K
 ARR recognized in Y1: \$150K (only H2 revenue)
 Churn: 0 (all in first year)

Year 2 (2027): Growth phase

YEAR 2 (2027) — BASE SCENARIO

Starting ARR: \$400K

New business:

Tier 1 new: +10 @ \$42K ACV avg = \$420K ARR new
 Tier 2 new: +7 @ \$180K ACV avg = \$1.26M ARR new
 Tier 3 new: +1 @ \$720K ACV = \$720K ARR new

Expansion (existing Tier 1 → Tier 2):

2 of 3 Tier 1 customers upsell to T2
 Upsell ARR: $2 \times (\$144K - \$36K) = \$216K$

Churn:

Tier 1: 1 churns (15% of 3 = 0.45, round to 0) → \$0
 Tier 2: 0 churns ($7\% \times 2 = 0.14$, round to 0) → \$0

Year 2 ARR end:

Starting: \$400K + New: \$2.4M + Expansion: \$216K - Churn: \$36K
 = \$2.98M $\approx$ \$3M ARR

Year 2 recognized revenue: $(\$400K + \$3M) / 2 = \$1.7M$

Year 3 (2028): Scale phase

YEAR 3 (2028) – BASE SCENARIO	
Starting ARR:	\$3M
New business:	
Tier 1 new:	+30 @ \$48K ACV = \$1.44M
Tier 2 new:	+20 @ \$210K ACV = \$4.2M
Tier 3 new:	+3 @ \$900K ACV = \$2.7M
Expansion:	
Tier 1 → T2:	5 customers @ \$150K expansion = \$750K
T2 upsell:	3 customers @ \$100K expansion = \$300K
Churn:	
Tier 1:	2 churns @ \$42K = -\$84K
Tier 2:	1 churns @ \$180K = -\$180K
Tier 3:	0 churns
Year 3 ARR end:	
$\$3M + \$8.34M + \$1.05M - \$264K = \$12.1M \approx \$12M \text{ ARR}$	
Year 3 recognized revenue: $(\$3M + \$12M) / 2 = \$7.5M$	

2.3. Summary revenue table

REVENUE SUMMARY (BASE SCENARIO)					
Metric	Y1	Y2	Y3	Y3 vs Y1	Note
ARR EOY	\$400K	\$3M	\$12M	30×	avg(start+end)
Recognized	\$150K	\$1.7M	\$7.5M	50×	
Customers	5	25	75	15×	
- Tier 1	3	13	38		
- Tier 2	2	10	30		NRR target
- Tier 3	0	2	7		
Net Rev Ret.	n/a	120%+	125%+		
Gross Margin	72%	70%	70%		

III. Unit economics

3.1. ARPU по тирам

	Year 1	Year 2	Year 3	
Tier 1 ARPU/year	\$36K	\$42K	\$48K	(+14% price increase)
Tier 2 ARPU/year	\$144K	\$180K	\$210K	(+20% upsell)
Tier 3 ARPU/year	n/a	\$720K	\$900K	(+25% expansion)
Blended ARPU:	\$80K	\$120K	\$160K	

3.2. Customer Acquisition Cost (CAC)

CAC Calculation:	
Sales & Marketing spend / новые customers в период	
Year 1 (founder-led sales, minimal spend):	
S&M spend:	\$120K (founder time + events + tools)
New customers:	5
CAC:	\$24K
Year 2 (first GTM hire + SDRs):	
S&M spend:	\$600K
New customers:	20
CAC:	\$30K
Year 3 (full GTM team):	
S&M spend:	\$1.8M
New customers:	53
CAC:	\$34K
Blended CAC payback period (at Gross Margin 70%):	
Year 2: CAC \$30K / (ARPU \$120K × 70%) = 4.3 months ✓ (target <12 months)	
Year 3: CAC \$34K / (ARPU \$160K × 70%) = 3.6 months ✓	

3.3. LTV (Life Time Value)

$LTV = ARPU \times \text{Gross Margin} / \text{Annual Churn Rate}$

По тирам:

Tier 1: $\$42K \times 0.75 / 0.15 = \$210K$ LTV

Tier 2: $\$180K \times 0.70 / 0.07 = \$1.8M$ LTV

Tier 3: $\$720K \times 0.65 / 0.05 = \$9.4M$ LTV

LTV:CAC ratio:

Tier 1: $\$210K / \$30K = 7:1$ ✓ (target >3:1)

Tier 2: $\$1.8M / \$30K = 60:1$ ✓

Tier 3: $\$9.4M / \$30K = 313:1$ ✓

Conclusion: unit economics very strong, especially for Tier 2+.

Tier 1 viable but less efficient; serves as top-of-funnel.

3.4. Gross margin waterfall

Revenue: \$100

Less: COGS

- Storage (S3 + CDN):	-\$3.50
- Validator compute (Tier 2 validations):	-\$5.20
- Audit log retention (S3 Glacier, 7yr):	-\$1.80
- Customer Success (0.25 FTE / \$10K customer):	-\$16.50
- Infra engineering (15% of infra cost):	-\$3.00

Total COGS:	-\$30.00
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Gross Profit:	\$70.00
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Gross Margin:	70%
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Notes:

- Customer Success cost dominates COGS (pure software companies target <10%)
- Reduces as automation increases (→ 75%+ at scale)
- Infrastructure cost is low and predictable
- No physical goods, licensing, or third-party royalties

IV. Operating expenses

4.1. Team plan (headcount by year)

HEADCOUNT PLAN			
Year 1 (2026):			
CEO/Founder (founder, not on payroll until funding):	1.0 FTE	\$0	
GTM Lead (hire Q4 2026 post-close):	1.0 FTE	\$80K	
Sr. Infra Engineer (hire Q4 2026):	1.0 FTE	\$90K	
Total hired Y1:	2 FTE (H2 only)		
Y1 payroll cost:	\$85K ($\frac{1}{2}$ -year × 2 people)		
Year 2 (2027):			
+ Research Lead (zkML / canonical hashing):	1.0 FTE	\$100K	
+ Crypto/Security Engineer:	1.0 FTE	\$95K	
+ Customer Success Manager:	1.0 FTE	\$70K	
+ 2nd Sales/BDR (MENA focus):	1.0 FTE	\$65K	
Total team Y2:	8 FTE (including Y1 + founder)		
Y2 payroll cost:	\$570K (annualized)		
Year 3 (2028):			
+ 3 more engineers (MCP, validator, storage):	3.0 FTE	\$300K	
+ 2 more AEs (Account Executives):	2.0 FTE	\$200K	
+ VP Engineering (fractional → full):	1.0 FTE	\$150K	
+ Legal / Finance (contractor → FTE):	1.0 FTE	\$90K	
Total team Y3:	15 FTE		
Y3 payroll cost:	\$1.31M (annualized)		

Geographic arbitrage note:

Engineering hires: Russia/Kazakhstan/Ukraine → ~50-60% of US equivalent

Sales hires: UAE/MENA → 20-30% premium over US for senior enterprise sales

Effective payroll blended discount: ~20% vs SF-equivalent team

4.2. OpEx waterfall

OPEX BREAKDOWN (BASE SCENARIO)		(\$K)		
Category	Year 1	Year 2	Year 3	
Payroll (engineering)	85K	285K	640K	
Payroll (sales/CS)	45K	285K	670K	
Founder salary	0K	120K	180K	
Total Payroll	130K	690K	1,490K	
Cloud infrastructure	24K	72K	240K	
SaaS tools (dev/ops)	18K	36K	72K	
Validator compute	6K	48K	180K	
Total Infrastructure	48K	156K	492K	
Sales & Marketing (events, ads, content)	75K	420K	1,320K	
R&D / Research grants (experiments, zkML POC)	30K	90K	180K	
G&A (legal, accounting, compliance, insurance)	40K	90K	150K	
TOTAL OPEX (excl. COGS)	323K	1,446K	3,632K	

4.3. P&L summary (base scenario)

P&L SUMMARY	Year 1	Year 2	Year 3
Revenue (recognized)	\$150K	\$1.7M	\$7.5M
COGS	-\$42K	-\$510K	-\$2.25M
Gross Profit	\$108K	\$1.19M	\$5.25M
Gross Margin %	72%	70%	70%
Payroll (non-COGS)	-\$130K	-\$690K	-\$1.49M
Sales & Marketing	-\$75K	-\$420K	-\$1.32M
Infrastructure (non-COGS)	-\$24K	-\$72K	-\$240K
R&D	-\$30K	-\$90K	-\$180K
G&A	-\$40K	-\$90K	-\$150K
Operating Income (EBIT)	-\$191K	-\$162K	\$1.87M
EBIT Margin	-127%	-10%	+24.9%
Funded by (pre-seed):	+\$700K		
Funded by (seed):	+\$4M		
Cash position EOY (* after seed round close Q1 2027)	+\$458K	+\$4.3M*	+\$6.2M*

Key milestones:
EBITDA breakeven: Q3 2028 (mid-Year 3)
First profitable quarter: Q4 2027 (on Gross Profit basis Q2 2027)

VI. Sensitivity analysis: три сценария

6.1. Bear case: медленная адопция, конкурентное давление

Bear Case Assumptions:

- Competitor (Anthropic internal team) releases similar product Q4 2026
- Pilot conversion rate 40% (vs 80% base)
- Churn 25% annual for Tier 1, 15% for Tier 2
- Sales cycles extend: +60 days each tier
- Pricing pressure: -30% on ACV due to competition

Bear Case Results:

Y1 ARR: \$100K (2 customers)
Y2 ARR: \$600K (8 customers)
Y3 ARR: \$2.5M (20 customers)

Y3 EBITDA: -\$800K (requires Series A to survive)

Mitigation: If competitor emerges → reposition as "open standard" vs "proprietary" → gain enterprise trust through openness.

Probability: 20%

Key risk: if Anthropic builds this natively, our value proposition shifts.
Thesis: we go CROSS-PROVIDER, Anthropic stays single-provider.
Even in bear case, CPP still works as neutral layer.

6.2. Base case (primary projection)

Детально описан в §§ II-V выше.

Base Case Assumptions:

- No major competitor for 18 months
- Pilot conversion rate 80%
- NRR 120%+ (expansion within accounts)
- Pricing holds

Probability: 60%

Y3 ARR: \$12M

EBITDA positive by Q3 2028

6.3. Bull case: viral adoption, enterprise fast-track

Bull Case Assumptions:

- EU AI Act enforcement creates regulatory push (Q4 2026)
- HuggingFace integrates CPP validator natively → distribution boost
- 2+ Tier 3 (sovereign AI lab) customers in Y1
- NRR 140%+ (rapid upsell within accounts)
- Pricing power: enterprise tier ACV 2× base

Bull Case Results:

Y1 ARR: \$500K (6 customers, including 1 T3)

Y2 ARR: \$5M (30 customers)

Y3 ARR: \$30M (100 customers, including 5 T3)

Y3 EBITDA: \$6M (20% margin)

Series A timing: Q1 2027 (pull forward)

Series A size: \$20-25M at \$100M+ valuation

Probability: 20%

6.4. Sensitivity table

Key variable: Pilot conversion rate (baseline: 80%)

Conv. Rate	Y1 ARR	Y2 ARR	Y3 ARR	Series A timing
40%	\$100K	\$700K	\$2.8M	Q2 2027 (delayed)
60%	\$220K	\$1.5M	\$7M	Q1 2027
80%	\$400K	\$3M	\$12M	Q1 2027 (base)
100%	\$550K	\$4.5M	\$18M	Q4 2026 (pull forward)

Key variable: ARPU growth (baseline: 15% annual)

ARPU growth	Y1 ARR	Y2 ARR	Y3 ARR	
0%	\$400K	\$2.3M	\$8.5M	
15%	\$400K	\$3M	\$12M	← base
30%	\$400K	\$3.5M	\$16M	
50%	\$400K	\$4.2M	\$22M	

VII. Comparison c public infrastructure companies

7.1. Comparable companies (exit multiples)

PUBLIC INFRA COMPARABLES (as of 2025-2026 market)				
Company	ARR at IPO	IPO Valuation	EV/ARR	Model
Datadog (2019)	\$100M	\$780M	7.8×	Usage-based SaaS infra
Snowflake (2020)	\$600M	\$33B	55×	Consumption data platform
HashiCorp (2021)	\$320M	\$3.1B	9.7×	Open-core infra tools
Grafana Labs (private)	\$200M+	\$6B	30×	Open-source observability
Stripe (private)	\$1B+	\$95B peak	~95×	Protocol infrastructure

CPP positioning:
Closest analogy: HashiCorp (open-core, infrastructure, developer-first)
Upside analogy: Stripe (protocol layer, network effects)

Conservative (HashiCorp multiple, 10× ARR):
Y3 ARR \$12M × 10 = \$120M valuation (Series A milestone)

Moderate (Datadog multiple, 15× ARR):
Y3 ARR \$12M × 15 = \$180M valuation

Optimistic (Snowflake-like if protocol wins, 40× ARR):
Y3 ARR \$12M × 40 = \$480M valuation (if canonical standard established)

7.2. Exit scenarios

Acqui-hire / Strategic acquisition scenarios:

Scenario A: Anthropic acquires CPP (2028-2029)

Rationale: integrate canonical encoding natively

Price: 15-25× ARR = \$180M - \$300M

Strategic value: eliminates competitor + gets protocol + team

Scenario B: HuggingFace acquires CPP

Rationale: add canonical infrastructure to their platform

Price: 20-30× ARR = \$240M - \$360M

Strategic value: complements their model hosting with context management

Scenario C: AWS / GCP / Azure acquires CPP

Rationale: differentiated cloud service for LLM enterprise customers

Price: 30-50× ARR = \$360M - \$600M

Strategic value: sticky enterprise contract mechanism

Scenario D: IPO (2030+)

Requires: ARR > \$50M, NRR > 120%, clear path to \$500M ARR

Timeline: if bull scenario plays out

Valuation: 20× ARR = \$1B+

VIII. KPI Dashboard для board reporting

8.1. Ключевые метрики

CPP BOARD DASHBOARD — MONTHLY UPDATE			
GROWTH			
ARR:	\$_____	MoM growth: _____%	vs plan: _____%
Customers:	_____	New this month: _____	Churn: _____
Pipeline:	\$_____ (weighted)		
RETENTION			
NRR (trailing 12m):	_____%	(target: >115%)	
Gross Churn:	_____%	(target: <8%)	
Expansion Revenue:	\$_____ (upsells + tier upgrades)		
UNIT ECONOMICS			
Gross Margin:	_____%	(target: 70%)	
CAC Payback:	_____ months	(target: <12)	
LTV:CAC ratio:	_____x	(target: >5x)	
CASH			
Cash on hand:	\$_____		
Net burn:	\$_____/mec		
Runway:	_____ months		
PRODUCT / TECHNICAL			
CIDs stored (total):	_____	(platform scale)	
Validations/day:	_____		
Uptime (30d):	_____%	(target: 99.9%)	
Cache hit rate avg:	_____%	(client-side metric)	
GTM / PIPELINE			
Retro analyses:	_____	(pipeline indicator)	
Demo calls:	_____	/mec	
LOIs pending:	_____		
VC conversations:	_____	(fundraising track)	

8.2. Leading indicators (сигналы в воронке)

Product-led growth signal:

- └─ cpp-cli analyze downloads/мес
- └─ GitHub stars velocity (cpp-core)
- └─ OSS contributor count

Sales pipeline signal:

- └─ Retro analyses completed (лагающий, но точный)
- └─ Shadow deployments active
- └─ Days to convert (retro → LOI → contract)

Market signal:

- └─ NRR trend (лучший indicator product-market fit)
- └─ Tier 1 → Tier 2 upgrade rate
- └─ Inbound requests (% of new pipeline vs outbound)

8.3. Investment timing по milestones

CAPITAL + MILESTONES ALIGNMENT

Pre-seed (\$700K) — Q4 2026

Milestone unlock:

- ✓ Anthropic cache alignment experiment results published
- ✓ 2+ pilot contracts signed (LOI acceptable)
- ✓ Reference implementation v1.0.0 stable
- ✓ 2 FTE hired

Seed (\$3-5M) — Q1 2027

Milestone unlock:

- ✓ \$250K ARR (3+ paying customers)
- ✓ 3 pilots with measurable results (% savings documented)
- ✓ arXiv preprint published + citations
- ✓ NRR > 100% (retention proven)
- ✓ 2nd enterprise customer signed

Series A (\$15-25M) — Q3 2027

Milestone unlock:

- ✓ \$2M ARR (20+ customers)
- ✓ NRR > 115%
- ✓ Gross Margin > 65%
- ✓ Enterprise Tier 2 customers paying >\$100K ACV
- ✓ Product working in 2+ geographies (US/EU + MENA or CIS)
- ✓ 1+ industry publication (NeurIPS workshop or similar)

IX. Приложение: детальные формулы

Формула оценки token savings (для sales)

```
monthly_savings($) =
  devs × sessions_per_day × days_per_month × turns_per_session ×
  context_tokens × repetition_rate × cache_discount ×
  token_price_per_M / 1_000_000
```

Например для команды 50 devs:

```
50 × 25 × 22 × 8 × 150_000 × 0.70 × 0.90 × 3 / 1_000_000
= 50 × 25 × 22 × 8 × 150000 × 0.63 × 3 / 1000000
= 50 × 25 × 22 × 8 × 283.5
= 50 × 25 × 22 × 2268
= 50 × 25 × 49896
= 50 × 1247400
= $62,370,000 ... / 1M (tokens)
→ $62,370 / мес
```

Sanity check: 50 devs × \$1,247 / dev / мес ≈ \$15K / мес savings.
(реалистично для shared codebase)

CPP revenue (20% rev share): \$15K × 0.20 = \$3,000/мес = \$36K ACV ← соответствует Tier 1

Формула CAC

```
CAC = (S&M_spend_in_period) / new_customers_in_period
```

```
Payback_months = CAC / (ARPU_monthly × gross_margin)
```

Rule of thumb:

```
Good SaaS: payback < 12 months
Great SaaS: payback < 6 months
CPP Y2: $30K / ($10K × 0.70) = 4.3 months → Great
```

Формула NRR

```
NRR = (ARR_start + expansion - contraction - churn) / ARR_start × 100%
```

Target >100%: каждый cohort растёт без новых customers.

Target >120%: category-leader level (Snowflake = 158% at IPO).

CPP path to 120% NRR:

```
Churn rate: 8% (1 - 0.08 = 0.92 retention)
Expansion: +30% (upsells + tier upgrades)
NRR: 0.92 × 1.30 = 1.20 = 120% ✓
```

Эта финансовая модель обновляется ежеквартально по мере накопления реальных данных. Контакт:
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